SOIL EROSION AND CONSERVATION · NOTE CM06

Non-Arable Land Measures — Diversion Drains, Contour Trenches, Half-Moon Terraces, Wattling and Geotextiles

Course topics: SWCE8.15 L6, SWCE8.15 L11, SWCE-510 T9

Non-arable lands

- Land unsuitable for conventional cropping because of steep slopes, erosion hazard, stoniness, shallow soils, wetness, flooding or extreme climate — or made so by road building and mining on steep hills. Vegetation is hard to establish because of heavy runoff and debris, lack of moisture and no fertile soil.
- They fall in land capability **Classes IV, V, VI, VII and VIII** — about 107 million ha of India's 328 million ha, often grouped as wastelands. Used without proper measures they erode severely and turn barren.
- Mechanical measures stabilise slopes, reduce erosion, hold debris and store moisture, and must be backed by vegetation. They are **slope-stabilisation measures** (diversion drains, contour trenching, contour wattling, retaining walls, crib structures, geotextiles) and **drainage-line treatments**.

Diversion drains

Laid across the slope to divert runoff from upstream areas away from vulnerable land and discharge it through a protected waterway at non-erosive velocity. Keep the bed slope within about 0.5% (non-erosive, non-silting). Broad, shallow drains are better than narrow, deep ones — they avoid silting and maintain themselves. Other details follow vegetated waterways.

Contour trenches

- Trenches dug along the contour, on a uniform level across hill slopes, with the spoil heaped on the downstream side as a ridge. Section usually trapezoidal (V-shaped when dug with V-ditchers), side slope 1 : 1, cross-section 1000 to 2500 cm² depending on soil depth; each catches the runoff from the strip above it.
- **Functions:** intercept runoff, conserve soil and water, raise soil moisture, break the slope length, reduce scour and sediment transport, and help vegetation establish. **Objectives:** (a) intercept runoff and store it in the soil profile; (b) moderate flash floods and improve groundwater recharge; (c) control erosion on denuded slopes.
- **Layouts:** continuous; short staggered; crescent-shaped (for tree planting); continuous for pasture development. Prevent silting with a stone barrier (~30 cm high) or a grass barrier on the upstream side; plant trees or shrubs on the downstream side or ridge.
- **Work best** in semi-arid and arid lands with intense rain, on fairly permeable soils, on denuded slopes, and for orchards, pastures and forests. Service life 4 to 5 years, by which time vegetation is established.
- **Soil benefits:** better aeration and infiltration, fines retained, more moisture, microbial activity and organic matter — better soil quality; the monsoon climate favours stable soil complexes. Keep depth and section, clean the trenches and harvest the grass.

Type	Features
Continuous	No break; 10 to 20 m long across the slope; for low rainfall, relatively flat land and mild storms. Sections 30 × 30 cm to 45 × 45 cm, with equalisers 20–25 cm wide at intervals to stop flow concentrating. Need high skill — improper construction breaches, poor alignment stores moisture unevenly, and a breach above can fail the trench below
Staggered	Preferred in medium rainfall on uniform or undulating land: alternate rows, 2 to 3 m long, rows 3 to 5 m apart — less prone to breaching
Graded	Continuous trenches with a gentle longitudinal slope that drain excess runoff at non-erosive velocity — for high rainfall (mean annual ≥ 1500 mm); level trenches suit low rainfall. They act as small drains: grade and section by Manning's formula so the trench holds intercepted runoff and passes the excess, without overflow, to a lower drain

Design

- Hold the runoff of a **4-year recurrence, 6-hour storm**: coarse soils store 60–70% of the estimated runoff, fine soils about 50%. Estimate runoff by the soil–cover complex (Curve Number) method.

$$Q = (P - 0.3S)^2/(P + 0.7S) \ (I_a = 0.3S; \text{ all soils except black soils at AMC II and III})$$
$$Q = (P - 0.1S)^2/(P + 0.9S) \ (\text{black soils, AMC II and III})$$
$$CN = 25400/(254 + S)$$

- **Section:** trapezoidal on hill slopes; triangular when the equipment requires it; square or rectangular on level land. Depth 40–50 cm in deep soils, 15–20 cm in shallow soils; area 1000–2500 cm² depending on soil, infiltration and rainfall. Convenient lengths: 3 m, 5 m, 7 m.

$$\text{Continuous trenches: } HI = A/Q \ (A = \text{cross-sectional area; } Q = \text{estimated runoff})$$
$$\text{Staggered trenches: } HI = (W \times D)/[1000 \cdot Q \cdot (1 + X/L)] \ (X = \text{gap between trenches, m; } L = \text{trench length, m})$$
$$\text{Vertical interval: } VI = (S/100) \times HI \ (S = \text{land slope, \%})$$

These equations count only the storage below ground level; net storage grows as water infiltrates. With W, D and Q all in mm, the staggered formula gives HI in m.

> **Illustration (not in the notes)**
> Staggered trenches 500 mm wide × 400 mm deep, design runoff Q = 20 mm, length L = 3 m, gap X = 2 m, slope 20%
> HI = 500 × 400 / [1000 × 20 × (1 + 2/3)] = 200000/33333 ≈ 6.0 m; VI = 0.20 × 6.0 = 1.2 m

Construction and maintenance

- Lay contours with a farm level or A-frame and mark trenches with a tape; work from the ridge downward.
- Heap spoil on the downstream side (keep some aside for refilling; use boulders or gravel as the toe of the mound). Finish before the monsoon: check grade, finish to design, build equalisers in continuous trenches with their tops at least 7.5 cm below the trench dike. Keep people and cattle out.
- Inspect periodically in early years and annually during or after the rains for breaches, rodent holes and failures. A trench serves until it fills up in 3–4 years.

Vegetation in trenched areas

- Aim: erosion control, more moisture and useful vegetation — silvi-pastoral systems are preferred. Tree spacing 4 to 6 m (plant to plant and row to row) in dry sub-tropical regions, 6 to 7 m in sub-humid regions. Fodder grass gives 2–3 harvests in the monsoon and 1–2 in winter; fodder trees are lopped 75–80% in winter.
- **Trench planting** (semi-arid and arid regions, limited depth, low rain; not black cotton, sodic or waterlogging-prone heavy soils): dig the trench, refill the bottom with topsoil leaving space for runoff, and plant seedlings after the monsoon starts, once soil is moist to 30 cm.
- **Mound planting** (humid and sub-humid regions): stabilise mounds with seedlings and grasses; broadcast fodder legumes such as *Stylosanthes hamata*.
- **Grasses:** *Cenchrus ciliaris*, *Dichanthium annulatum*, *Eulaliopsis binata*, *Chrysopogon fulvus*, *Lasiurus sindicus*. **Trees:** babool (*Acacia nilotica*), kikar (*Parkinsonia aculeata*), neem (*Azadirachta indica*), khair (*Acacia catechu*), shisham (*Dalbergia sissoo*), *Prosopis juliflora*, jatropha (*Jatropha curcas*), subabul (*Leucaena leucocephala*). Plant ~1-year-old seedlings in the monsoon after a good soaking rain; avoid waterlogged soils.
- **Limitations:** trenches interfere with continuous farming and do not suit very shallow soils or impermeable layers near the surface, very coarse soils with little clay or organic matter, shrink-swell clays, stony soils, areas with a history of slumping, places where runoff from above exceeds trench capacity, very steep land, humid heavy-rainfall areas (uneconomic), or areas with seepage or inflow from perennial streams.

Half-moon terraces (crescent trenches)

- Individual semi-circular basins, 0.75 m to 1.5 m in diameter (adjustable), with a shoulder bund on the lower edge — for fruit and other tree crops and semi-permanent or permanent crops on steep slopes, conserving moisture and checking erosion.
- Suit sloping, uneven, dissected land with shallow to moderately deep soils and low to moderate rainfall. Cut the uphill side in a semi-circle, fill the soil on the downhill side, give the basin an inward slope and protect it with a bund (riser slope 0.75 : 1). In humid regions add a small outlet and ditch for excess water. Plant the sapling in the centre.

Contour wattling

- Mechanical stabilisation of steep slopes plus vegetation — for deep gullies, landslides, roadside hill slopes and mine spoils on slopes up to about 33%. Wattles every 3–5 m break the slope into short lengths.
- **Construction:** dig a contour trench 30 cm wide and about 1 m deep; drive wooden posts 1.25 to 1.5 m long and 8–12 cm thick, 1–1.25 m apart; fill with brushwood bundles (5–6 cm thick) and weave brushwood around the posts up to 30 cm above ground.
- **Live wattling:** branches 8–12 cm thick and 1.25–1.5 m long are planted in the trench and sprout into trees — e.g. *Ficus infectoria*, *Lannea coromandelica*, *Erythrina suberosa*, *Vitex negundo*, *Salix tetrasperma* (willow), *Hamiltonia suaveolens*.
- **Mulch** the inter-spaces with 5–10 cm of grass — about 14,000 kg of air-dry grass per hectare — to conserve moisture and give a congenial microclimate.
- Not suitable on very loose or powdery rock or on slopes steeper than 66%.

Geotextiles

- Woven nets of natural fibre (jute or coir) that stabilise steep slopes — landslide-prone areas, mine spoils, steep road cuttings. The fibre degrades in about 2 years, by when vegetation has established.
- **Specifications:** open mesh 25 × 25 mm; geojute net 2–5 mm thick with 10 mm apertures. Effective on highly degraded land and Himalayan border roads — 60–70% effective for initial vegetation establishment.

1. Provide a diversion drain above the problem area; build gully plugs and stone barriers if needed.
2. Dig planting pits and an anchor trench at the upper end.
3. Spread the net, anchor its upper end in the trench and backfill; overlap adjoining nets (22 cm overlap for 1 m effective width).
4. Drive wooden stakes 1.5 m long — 1 m below ground, 50 cm above.
5. Plant rooted slips or bush cuttings in the openings — kans (*Saccharum spontaneum*), broom grass (*Thysanolaena maxima*), *Ipomoea*, *Vitex negundo*, *Arundo donax*, giant napier.
6. Build a masonry toe wall at the lower end if needed.

Trench cultivation ("underground greenhouse")

- For cold deserts such as Ladakh, where temperatures stay near or below freezing most of the year and polyhouses are expensive. Trenches trap soil and solar heat and stay warmer, letting farmers raise vegetables and flowers cheaply; 2 or more vegetable crops per season have been grown in Ladakh.
- **Site:** plenty of sunlight, east–west orientation. **Size:** 3–10 m long (longer trenches lose temperature), 2–4 m wide, 60–125 cm deep, about 50 cm apart. Scrape and keep the fertile topsoil, dig to depth and heap soil between trenches; line side walls with stone if deeper than 60 cm.
- **Cover:** UV-stabilised LDPE sheet (200 micron) on mild-steel, GI-pipe or bamboo supports, with 150-gauge black polyethylene at night.
- **Crops:** leafy vegetables in winter; nurseries of cabbage, capsicum and cauliflower in spring; cucurbits in summer.
- Not for low-lying land or where a high water table causes seepage; ensure drainage first.

Species names are given in their standard spelling (the notes have Stylosanthes hamatai, Lasirus, Jatropha curcus, Dalbergia sisso, Cenisrema and Dolicos for S. hamata, Lasiurus, J. curcas, D. sissoo, Centrosema and Dolichos).

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. ICAR–Indian Institute of Soil and Water Conservation, Dehradun (contour trenching, geo-jute and wattling practices). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Area of non-arable land in India is about

(a) 37 Mha (b) 68 Mha
(c) 107 Mha (d) 173 Mha

Q2. Bed slope of diversion drains is usually kept within

(a) 0.05% (b) 0.5%
(c) 2% (d) 5%

Q3. Cross-sectional area of contour trenches is usually

(a) 100–250 cm² (b) 1000–2500 cm²
(c) 1–2.5 m² (d) 5 m²

Q4. Staggered contour trenches are typically

(a) 0.5–1 m long (b) 2–3 m long, rows 3–5 m apart
(c) 10–20 m long (d) 50 m long

Q5. Graded trenches suit areas with mean annual rainfall of at least

(a) 500 mm (b) 800 mm
(c) 1000 mm (d) 1500 mm

Q6. Contour trenches are designed to hold runoff from a storm of

(a) 2-yr, 1-h (b) 4-yr, 6-h
(c) 10-yr, 24-h (d) 25-yr, 6-h

Q7. Equaliser tops in continuous trenches are kept at least ___ below the trench dike

(a) 2.5 cm (b) 7.5 cm
(c) 15 cm (d) 30 cm

Q8. Contour wattling is not effective on slopes steeper than

(a) 15% (b) 33%
(c) 50% (d) 66%

Q9. Geojute nets degrade in about

(a) 3 months (b) 2 years
(c) 5 years (d) 10 years

Q10. Trench cultivation in Ladakh uses trenches about

(a) 30 cm deep (b) 60–125 cm deep
(c) 2–3 m deep (d) 5 m deep

Answers

Q1 — (c) 107 Mha
Q2 — (b) 0.5%
Q3 — (b) 1000–2500 cm²
Q4 — (b) 2–3 m long, rows 3–5 m apart
Q5 — (d) 1500 mm
Q6 — (b) 4-yr, 6-h
Q7 — (b) 7.5 cm
Q8 — (d) 66%
Q9 — (b) 2 years
Q10 — (b) 60–125 cm deep